Teach, Check, Respond

Session 1 — Find the Whole · Session 2 — Solve for the Whole · Maryland 6.AT.4

LESSON ROUTE

about 45 minutes per session

Warm up · 5 min	Warm-Up page (Session 1). Note who needs the prerequisite sheet.
Launch · 7 min	Slides: name the target. Students predict an answer to the Model, Reason, Explain task.
Model · 10 min	Guided Notes page for the session. Draw the deck's model next to the worked example.
Practice · 18 min	Practice Form A: core #1, #2, #4, #6 and Find the Error. Add more from what you see.
Check · 5 min	Exit Ticket. Sort into got it / almost / need help with the table below.

Shorter period: keep the model and the exit ticket; trim practice. Form B is for reteach or retake, so keep it unseen until then.

SESSION 1 · FIND THE WHOLE

I can find the whole when I know a part and the percent it represents.

EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

#	Expected answer	Next prompt if it is not there yet
1	a. 110 b. 36	11 is 10% and 9 is 25%. Why do you multiply one by 10 and the other by 4?
2	10% = 28 ÷ 4 = 7; 100% = 70 games	Show 28 games as 40% on a 10-part bar. What does one part stand for?
3	**C** (greater than the part)	Make up a part that is 50% of a whole. Is your whole more or less than the part? Explain.

THE ERROR TO WATCH FOR

Treats the known part as the whole or multiplies it by the percent (0.3 × 27 = $8.10), so the answer is smaller than the part.

Reteach Have the student write $27 under the shaded 30% section and ask whether the end of the bar (100%) must be more or less than $27 before any computing.

- Multiplying the part by the percent (0.8 × 36) instead of scaling up, giving a whole smaller than the part.
- Dividing by the percent as a whole number (36 ÷ 80) instead of stepping down to 10%.

Calculation slip only: use Session 1 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

Check	A library has lent out 43 laptops. That is 20% of all its laptops. How many laptops does the library have?
Answer	10% = 43 ÷ 2 = 21.5; 100% = 21.5 × 10 = 215 laptops (or 43 × 5 = 215). Check: 0.20 × 215 = 43.

SCORE THE CHECK

3	Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time.
2	Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem.
1	Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page.
0	Not enough to tell yet. Ask the student to point, draw or say it before writing.

LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "The part is ___. It is ___% of the whole." Partner B: "So 10% is ___, and 100% is ___." Then switch."), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.

SESSION 2 · SOLVE FOR THE WHOLE

I can write and solve an equation to find the whole from a part and a percent.

EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

#	Expected answer	Next prompt if it is not there yet
1	a. 52 b. 70	Solve 0.25 × w = 13 another way: 25% is one fourth, so what do you multiply 13 by?
2	a. 0.6 × w = 138 b. w = 138 ÷ 0.6 = $230	Check the bike: what is 60% of your first price? Is it $138?
3	**B** (40)	Someone chose 6.4 for 16 = 0.4w. What did they do, and how can you tell it is wrong?

THE ERROR TO WATCH FOR

Writes the percent as a whole number (45 × w = 9) or multiplies the part by the decimal (w = 0.45 × 9 = 4.05) instead of dividing the part by the decimal.

Reteach Point to the bar: the 9 km is the shaded 45% and w is the whole bar, so write 0.45 × w = 9 and ask what undoes multiplying by 0.45.

- Multiplying the part by the percent (w = 0.25 × 52) instead of dividing.
- Dividing by the percent number (1,050 ÷ 35) instead of the decimal (1,050 ÷ 0.35).

Calculation slip only: use Session 2 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

Check	A team has sold 87 T-shirts. That is 75% of its goal. Write and solve an equation for the goal.
Answer	0.75 × g = 87; g = 87 ÷ 0.75 = 116 T-shirts. Check: 0.75 × 116 = 87.

SCORE THE CHECK

3	Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time.
2	Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem.
1	Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page.
0	Not enough to tell yet. Ask the student to point, draw or say it before writing.

LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "The part is ___ and the percent is ___." Partner B writes ___ × *w* = ___ and solves. Both check: ___ × ___ = ___."), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.